

Math 372: Homework 05

Due Friday, September 25

Problem 1 (Binomial random variable). Let $\alpha \in (0, 1)$ be a fixed parameter. Suppose we flip a coin 4 times, each time with the probability of heads being α . Let

$X =$ (the number of heads out of the 4 flips)

(a) Make a table like the following, including all 16 possible outcomes:

Outcome	Value of X	Probability
HHHH	4	α^4
HHHT	3	$\alpha^3(1 - \alpha)$
HHTH	3	$\alpha^3(1 - \alpha)$
HHTT	2	$\alpha^2(1 - \alpha)^2$
...	...	...

(b) From now on, let's assume that $\alpha = \frac{1}{2}$. In other words, we are flipping fair coins. Recalling that $p(x) = \mathbb{P}[X = x]$, fill out a table like the following:

x	0	1	2	3	4
$p(x)$					

(c) Verify that $\sum_x p(x) = 1$.

(d) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

(e) Use your answer from the previous part to compute $\text{Var}(X)$.

Problem 2. A new streaming service has exactly 6 shows, numbered 1 through 6. Suppose we randomly select a subscriber and ask which shows they watch. For each $i = 1, 2, \dots, 6$, let $E_i = \{i\}$ be the event that the subscriber watches show i . Suppose that

$$P(E_1) = P(E_6) = .10, \quad P(E_2) = P(E_5) = .15, \quad P(E_3) = P(E_4) = .25.$$

Define events A, B, C by

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad C = \{2, 3, 4, 5\}, \quad D = \{1, 3, 5\}.$$

Recall that

Events A and B are **independent** if and only if $\mathbb{P}[A \cap B] = \mathbb{P}[A] \mathbb{P}[B]$

(a) Are A and B dependent or independent?

(b) Are A and C dependent or independent?

(c) Are A and D dependent?

(d) Can two mutually exclusive events A and B with $P(A) > 0$ and $P(B) > 0$ be independent? Why or why not?

Problem 3. Assume that A and B are independent. Explain why A^c and B must be independent, and also why A^c and B^c must be independent.

Problem 4 (Exercise 2.5.77). During a long siege, five explosive devices are placed underneath the walls of the enemy's fortress, and are set to detonate simultaneously. Sappers report that each bomb has a 4% chance of failing to detonate at the specified time. Assuming that the probabilities are independent, calculate $\mathbb{P}[\text{at least one detonation occurs}]$ and $\mathbb{P}[\text{at least one bomb fails to detonate}]$

Problem 5 (Exercise 2.5.71). A space mining company currently has two active projects, one on the moon and one in the Kuiper belt. Let L be the event that the lunar mining project is successful, and let K be the event that the Kuiper belt mining project is successful. Suppose that L and K are independent events with $\mathbb{P}[L] = 0.4$ and $\mathbb{P}[K] = 0.7$.

- (a) If the lunar project is not successful, what is the probability that the Kuiper belt project is also not successful? Explain your reasoning.
- (b) What is the probability that at least one of the two mining projects will be successful?
- (c) Given that at least one of the two projects is successful, what is the probability that only the lunar project is successful?

Problem 6 (Similar to Example 2.31). An email provider observes that 50% of emails are spam. It intends to filter spam emails before they reach the inbox. The filter's accuracy for detecting spam email is 99% and chances of tagging a non-spam email as spam is 5%.

- (a) Draw a tree diagram to illustrate this situation
- (b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}[\text{email is not spam} \mid \text{email is tagged as spam}]$$

Problem 7 (Rolling two dice). Two 6-sided dice are rolled and added together. Let X be the sum.

- (a) What is the probability mass function of X ? [*Hint*: First determine $p(1), p(2), \dots$ and so on.]
- (b) Determine the cumulative density function of X and graph it.
- (c) What is the expected value of X ?

Problem 8 (Rolling a d6 'with advantage'). In the game *Dungeons & Dragons*, there is a dice-rolling mechanic called "rolling with advantage" in which the player gets to roll a dice twice and keep the largest value. This bonus is typically regarded as a good thing, as higher rolls tend to be better in the game. In this problem, we'll investigate how significant this bonus is. Two 6-sided dice are rolled. Let M be the maximum of the two rolls.

- (a) What is the probability mass function of M ? [*Hint*: First determine $p(1), p(2), \dots$ and so on.]
- (b) Determine the cumulative density function of M and graph it.
- (c) What is the expected value of M ?
- (d) Suppose you are given the choice between two magical weapons: a magical sword that deals $1d6 + 1$ damage, and a magical spear that deals $1d6$ damage *with advantage*. Which magical weapons has a higher expected damage? (*Here, 1d6 means "roll one six-sided dice"*.)

Problem 9 (2.4.63). For customers purchasing a refrigerator at a certain appliance store, let A be the event that the refrigerator was manufactured in the U.S., B be the event that the refrigerator had an icemaker, and C be the event that the customer purchased an extended warranty.

Relevant probabilities are

$$\begin{aligned}\mathbb{P}[A] &= .75 & \mathbb{P}[B \mid A] &= .9 & \mathbb{P}[B \mid A^c] &= .8 \\ \mathbb{P}[C \mid A \cap B] &= .8 & \mathbb{P}[C \mid A \cap B^c] &= .6 \\ \mathbb{P}[C \mid A^c \cap B] &= .7 & \mathbb{P}[C \mid A^c \cap B^c] &= .3.\end{aligned}$$

- (a) Construct a tree diagram consisting of first-, second-, and third-generation branches, and place an event label and appropriate probability next to each branch.
- (b) Compute $\mathbb{P}[A \cap B \cap C]$.

- (c) Compute $\mathbb{P}[B \cap C]$.
- (d) Compute $\mathbb{P}[C]$.
- (e) Compute $\mathbb{P}[A \mid B \cap C]$, the probability of a U.S. purchase given that an icemaker and extended warranty are also purchased.

Problem 10. Two cards are randomly drawn from a deck of cards. Let

$A = [\text{at least one ace is drawn}]$

$B = [\text{both cards are aces}]$

$C = [\text{the ace of clubs is drawn}]$

Compute the following conditional probabilities:

- (a) $\mathbb{P}[C \mid B]$
- (b) $\mathbb{P}[B \mid C]$
- (c) $\mathbb{P}[B \mid A]$
- (d) Note that the answer to (b) is almost twice as large as the answer to (c). Do you find this result surprising?

Problem 11. A 5-card poker hand is a **full house** if it has a 3-of-a-kind and 2-of-a-kind.

- (a) How many possible full houses are there?
- (b) Suppose you are dealt a random poker hand. What's the probability that you get a full house?